

		fringe separation.
16	D	Using single slit diffraction, $\sin \theta = \frac{\lambda}{b}$ $\sin \theta = \frac{600 \times 10^{-9}}{0.20 \times 10^{-3}}$ $\theta = 0.0030 \text{ rad}$ $s = r\theta = 2.0 \times 0.0030 = 0.0060 \text{ m}$ width of central maximum $\approx 2s = 2 \times 0.0060 = 0.012 \text{ m} = 12 \text{ mm}$
17	C	Since Q experiences no net force